



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

PHISHING AWARENESS AND SIMULATION

The domain of the Project:
CYBERSECURITY

Mentor (and their designation):

Mr. Derick Mathew Johnson
Trainer, Sure ProEd

Member:

Ms. P.Harshitha

Period of the project :

February 2026 to August 2026



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Declaration

The project titled “Phishing Awareness and Simulation” and has been mentored by Mr. Derick Mathew Johnson, organised by SURE Trust, from February 2026 to August 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Member:

Ms. P.Harshitha

Signature

Mentor's Name :

Mr. Derick Mathew Johnson

Trainer, Sure ProEd

Prof. Radhakumari

Executive Director & Founder

SURE Trust



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Executive Summary

Objective

Phishing attacks are among the most common cybersecurity threats, targeting users through deceptive emails and social engineering techniques. The objective of **Phishing Awareness and Simulation** is to provide organizations, educational institutions, and enterprises with a permission-based phishing awareness simulation platform that helps evaluate and improve users' ability to recognize phishing attacks. The platform enables administrators to conduct ethical phishing simulations, monitor participant behaviour, provide basic phishing awareness training, and measure improvements through before-and-after campaign analysis.

Project Methodology

Phishing Awareness and Simulation was developed using **Python, Flask, SQLite, HTML, CSS, JavaScript, Chart.js, and GoPhish with SMTP integration**. The platform follows a structured workflow that includes volunteer registration, informed consent, phishing campaign creation, email simulation, participant activity tracking, awareness training, quiz assessment, and report generation. Administrators can create phishing campaigns using multiple phishing email templates, launch campaigns through GoPhish or SMTP, and monitor campaign progress through a centralized dashboard.

The platform also includes a **basic self-paced awareness training module** that introduces participants to common phishing attacks such as email phishing, spear phishing, smishing, vishing, whaling, and social engineering. After completing the training, participants take a short quiz to assess their understanding before participating in the second phishing simulation. During simulations, the system records email clicks, simulated credential submission attempts, phishing reports, and quiz scores without storing real passwords or sensitive user credentials, ensuring ethical and secure testing.



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Key Findings

The developed platform successfully demonstrates a complete phishing awareness workflow, from participant registration and consent management to phishing simulation, awareness training, quiz assessment, and comparative reporting. The administrative dashboard provides visual reports that compare participant behaviour before and after awareness activities using metrics such as click-through rate, simulated credential submission rate, phishing reporting rate, average reporting time, and quiz performance. The project shows that organizations can use phishing simulations and awareness activities to identify users who are more susceptible to phishing attacks and evaluate the effectiveness of their awareness initiatives through measurable behavioural data.

Recommendations

The current implementation successfully supports participant registration, consent management, phishing campaign management, volunteer dashboards, awareness training, quiz assessment, campaign tracking, and analytical reporting. Future enhancements may include interactive training content, scenario-based exercises, automated campaign scheduling, role-based access control, support for multiple email providers, and advanced analytics for more detailed behavioural assessment.

Significance for Executives and Stakeholders

Phishing Awareness and simulation provides organizations with a centralized platform for conducting ethical phishing awareness campaigns while monitoring participant behaviour through measurable reports and dashboards. The platform enables administrators to identify awareness gaps, evaluate campaign effectiveness, and make informed decisions regarding future cybersecurity awareness programs. By integrating participant registration, consent management, phishing simulations, basic awareness training, quizzes, campaign monitoring, and reporting within a single web application, the project offers a practical and cost-effective solution for improving phishing awareness and strengthening the human aspect of cybersecurity.



Introduction

A web-based phishing awareness and simulation platform developed to help organizations, educational institutions, and training environments conduct controlled phishing awareness campaigns in an ethical and measurable manner. The platform combines participant registration, informed consent, phishing campaign management, awareness training, behavioural tracking, and reporting into a single web application.

Project Context

The platform is intended for **controlled and permission-based awareness programs**.

Before any phishing simulation is conducted, participants must voluntarily register on the platform and accept the digital consent agreement. Only participants who have provided consent become eligible for phishing campaigns.

After registration, administrators create phishing campaigns by selecting:

- Campaign Name
- Campaign Round
- Phishing Email Template
- Consented Volunteers

The platform first attempts to launch campaigns through **GoPhish**. If GoPhish is unavailable, the application automatically switches to the configured SMTP service to maintain uninterrupted campaign execution.

Problem Statement

The project addresses the need for a centralized platform that enables organizations to conduct ethical phishing awareness campaigns while measuring participant behaviour throughout the awareness process.

Instead of relying on independent tools for campaign creation, participant management, awareness training, quizzes, and reporting, the proposed



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platform integrates these components into a single web application.

Goals

- Develop a centralized phishing awareness platform. Register volunteer participants and obtain informed consent before simulations.
- Create phishing campaigns using multiple phishing email templates. Launch campaigns through GoPhish with SMTP fallback support.
- Track participant interactions such as email clicks, simulated credential submissions, and phishing reports.
- Provide a basic awareness training module covering common phishing techniques. Conduct quiz-based assessment after training.
- Compare participant behaviour before and after awareness activities. Generate graphical reports and recommendations for administrators.

Scope

- Create simulated phishing email templates (in a controlled, permission-based lab). Track click-through rates and run a short awareness module. Generate a report on results and improvement. Study phishing techniques (spear phishing, pretexting, urgency cues) and design consent forms for the test group.
- Set up GoPhish or a Flask-based campaign server with at least five phishing email templates covering different pretexts. Build simulation landing pages that track clicks and credential submissions without storing real user data.
- Run a baseline phishing campaign against a volunteer test group and collect click rates and time-to-report metrics. Build a self-paced web-based training module with quizzes explaining how to identify phishing attempts.



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- Run a second campaign after training and compare before/after metrics across all tracked behaviors. Write a full awareness campaign report with statistical analysis of behavior change and concrete improvement recommendations.

Tools & Tech: Python/Flask or GoPhish, HTML/CSS (landing pages), Chart.js(reports)

Outcome: Simulation platform + training module + before/after metrics report + awareness recommendations

Limitations

Although the platform successfully implements the core functionality of a phishing awareness simulation system, the current version has limitations such as automated campaign scheduling, and AI-based behavioural analysis are not included in the current implementation.

Innovation component

A **single web-based platform** which introduces a **consent-first approach** and instead of relying on separate applications for campaign management, participant registration, awareness training, and reporting.



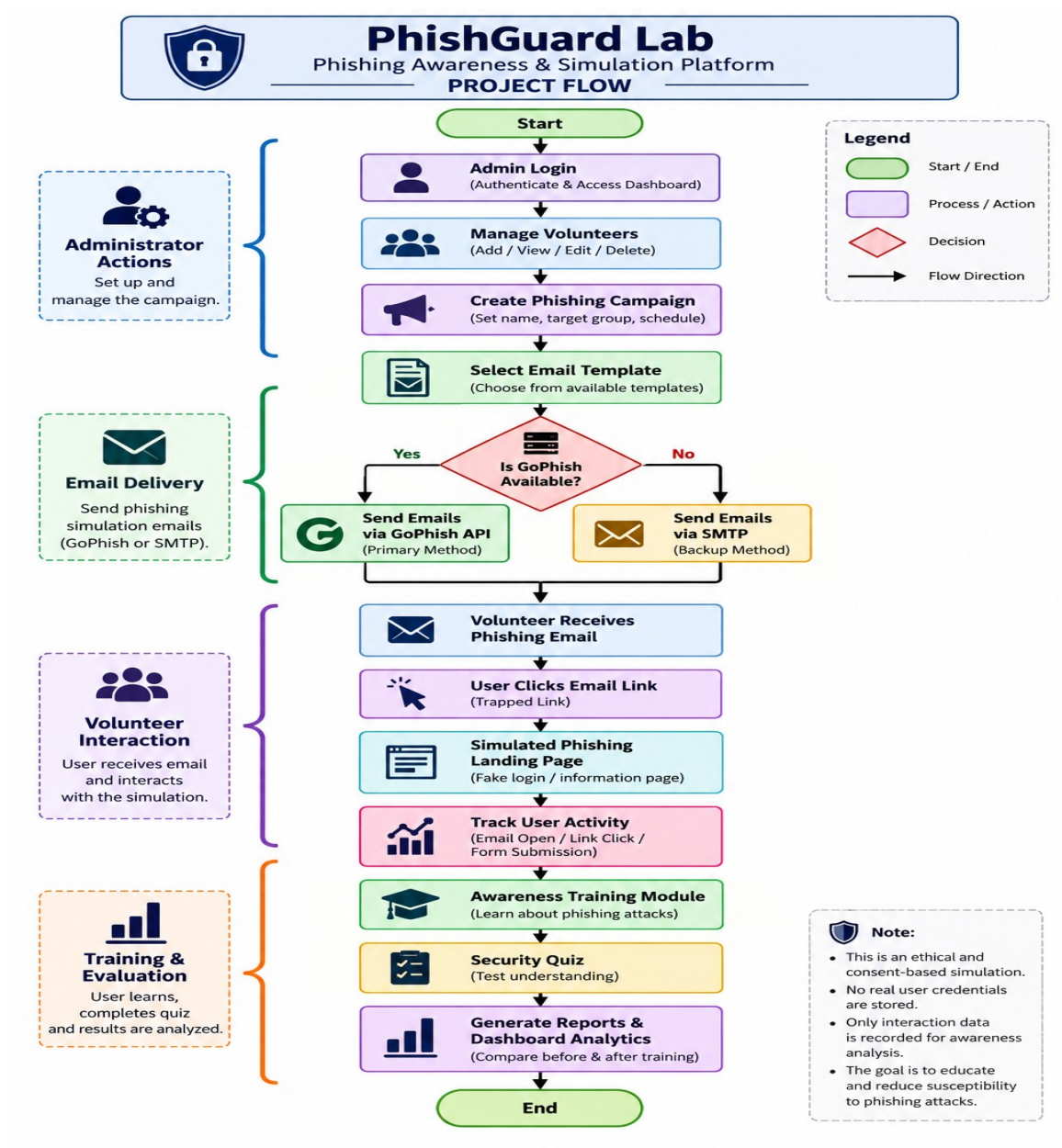
Project Objectives

The main objectives of the project are:

1. To develop a secure web-based platform for conducting ethical phishing awareness simulations.
2. To educate users about phishing attacks through interactive training modules and quizzes.
3. To simulate real-world phishing emails in a controlled environment to evaluate user awareness.
4. To track participant activities such as email opens, link clicks, and form submissions for analysis.
5. To compare users' awareness levels before and after training using measurable metrics.
6. To provide administrators with a centralized dashboard for managing volunteers, campaigns, and reports.
7. To support both **GoPhish integration** and **SMTP-based email delivery** so campaigns can continue even if GoPhish is unavailable.
8. To generate analytical reports and graphical dashboards that help organizations assess cybersecurity awareness and identify areas for improvement.



Project Flow



System Architecture



Methodology and Results

Methods / Technology Used

The **Phishing Awareness & Simulation Platform (PhishGuard Lab)** was developed using a client-server architecture to provide an ethical environment for phishing awareness training. The project follows a systematic workflow where an administrator creates and manages phishing campaigns, while volunteers participate in simulations and awareness activities.

The administrator first logs into the platform using secure authentication. After logging in, volunteers are registered and managed through the dashboard. The administrator then creates phishing campaigns by selecting phishing email templates and configuring campaign details. The platform primarily uses **GoPhish** for sending phishing simulation emails. If GoPhish is unavailable or offline, the platform automatically switches to **SMTP** as a backup email delivery service to ensure uninterrupted campaign execution.

When volunteers receive phishing emails and click on the embedded links, they are redirected to simulated phishing landing pages that imitate legitimate websites. The platform records user interaction data such as email delivery, email opens, link clicks, and simulated form submissions. No real passwords or confidential information are stored during the simulation.

After completing the phishing simulation, volunteers are redirected to a cybersecurity awareness training module containing educational material on phishing attacks, social engineering techniques, email security, and safe browsing practices. Users then complete an online quiz to evaluate their understanding of phishing concepts.

Finally, the administrator can access a dashboard that displays reports, charts, and statistics comparing user awareness before and after training. These reports help evaluate the effectiveness of the awareness program and identify areas requiring additional training.



Tools/Software used

The project was developed using the following technologies:

Tool / Technology	Purpose
Python	Backend programming language
Flask	Web application framework
HTML5	Structure of web pages
CSS3	User interface styling
JavaScript	Interactive frontend functionality
Bootstrap	Responsive user interface
SQLite	Database for storing application data
GoPhish	Phishing campaign management
SMTP	Backup email delivery service
Chart.js	Dashboard charts and analytics
Git & GitHub	Version control and project management
Visual Studio Code	Code editor
Ubuntu Linux	Development environment

Data collection approach

The project collects only the information necessary to evaluate cybersecurity awareness. During phishing simulations, the platform records campaign-related activities such as email delivery status, email open status, link click events, simulated form submissions, quiz scores, training completion status, and campaign participation.

The collected information is stored securely in the SQLite database and is used only for generating awareness reports and performance analytics. To ensure ethical operation, the platform never stores actual passwords or sensitive personal information entered on simulated phishing pages. Instead, only user



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interaction events are recorded for educational analysis.

Project Architecture

The architecture of the project consists of multiple integrated components that work together to provide phishing awareness training.

Administrator Module

Provides secure authentication and allows administrators to manage volunteers, create phishing campaigns, configure email templates, monitor campaign progress, and view reports.

Volunteer Module

Allows participants to receive phishing simulation emails, access simulated landing pages, complete cybersecurity awareness training, and take quizzes.

Flask Backend

Acts as the central controller of the application. It processes requests from users, communicates with the database, manages campaign information, interacts with GoPhish through its REST API, and automatically switches to SMTP whenever GoPhish is unavailable.

SQLite Database

Stores administrator information, volunteer records, campaign details, quiz results, interaction logs, and training completion records.

Email Delivery Module

Sends phishing simulation emails. GoPhish serves as the primary email engine, while SMTP functions as a backup email service to ensure reliable campaign delivery.

Reporting Module

Analyzes user interaction data and generates visual dashboards showing campaign success rates, click rates, quiz scores, and awareness improvements.



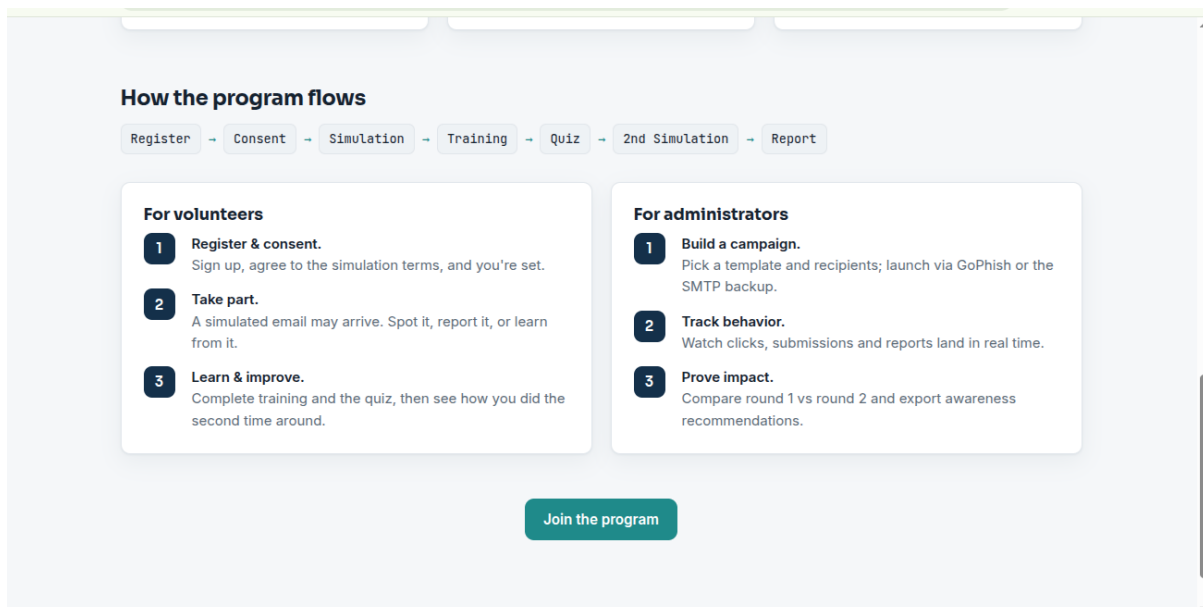
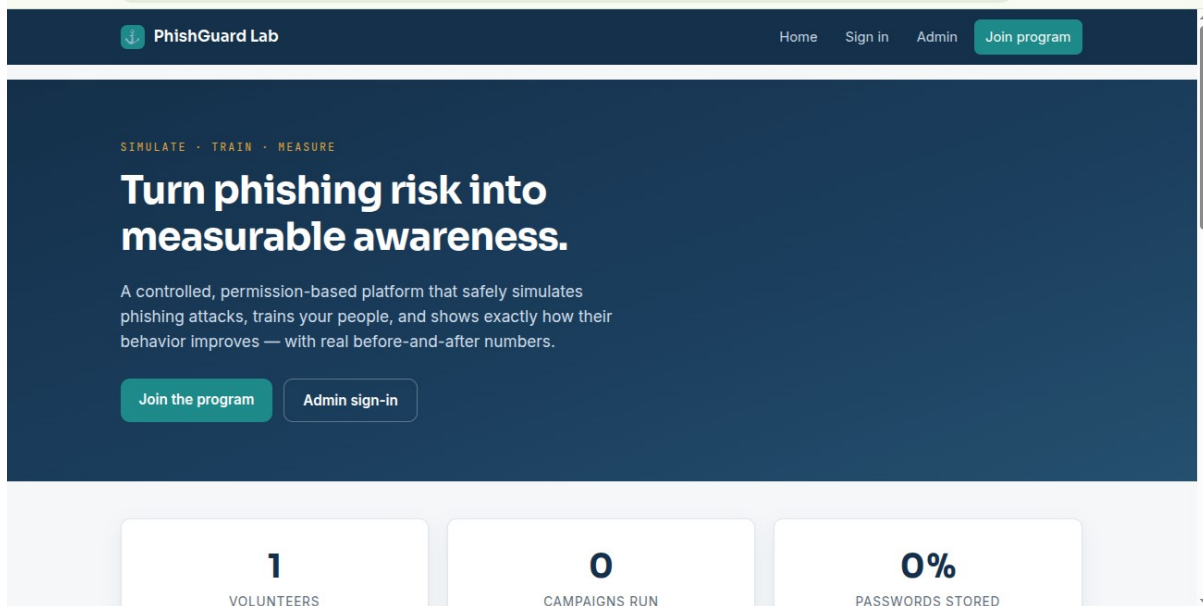
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Final project working screenshots

The completed project consists of several functional modules. Screenshots of each module should be inserted below along with a brief explanation.

Home Page

Displays project introduction and navigation options.





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Administrator Login

Provides secure access to the administration dashboard.

The screenshot shows the 'Administrator sign-in' page of the PhishGuard Lab. The page has a dark blue header with the PhishGuard Lab logo and navigation links: Home, Sign in, Admin, and a 'Join program' button. The main content area is light gray and features a sign-in form with fields for 'Username' (containing 'admin') and 'Password' (masked with dots). A 'Sign in' button is at the bottom of the form. Below the form, a footer bar contains the text: 'PhishGuard Lab · A consent-based phishing awareness & simulation platform for training and research. Simulated emails only. Passwords are never stored.'

The screenshot shows the 'Campaign overview' dashboard of the PhishGuard Lab. The page has a dark blue header with the PhishGuard Lab logo and navigation links: Overview, Campaigns, Reports, and Sign out. The main content area is light gray and features a 'Campaign overview' section. At the top right of this section, it says 'Phishing engine: GoPhish offline - SMTP backup ready'. Below this, there are four cards showing statistics: '1 VOLUNTEERS', '1 CONSENTED', '0 CAMPAIGNS', and '0 QUIZZES TAKEN'. Below these cards are three links: 'Campaigns →', 'Volunteers →', and 'Reports →', each with a brief description. At the bottom, there is a 'Recent campaigns' section with the text 'No campaigns yet. Create one.'



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Volunteer Consent Form

Full name
Harshitha G15-CS

Email address
harshithapeddamuregs15@gmail.com
Simulated emails and account sign-in use this address.

Create an account password

This is your platform login — separate from anything in a simulation.

Participant type
Employee

Organization

Department

[Continue to consent](#)

PhishGuard Lab Home Sign in Admin [Join program](#)

Volunteer sign-in

Email

Password

[Sign in](#)

[New here? Join the program](#)

PhishGuard Lab · A consent-based phishing awareness & simulation platform for training and research. Simulated emails only. Passwords are never stored.

Campaign Creation

Used to configure phishing simulation campaigns.



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PhishGuard Lab

OverviewCampaignsReportsSign out

Campaigns

Create a campaign

Campaign name

Round

Round 1 — baseline (before training) ▾

Email template

Password Reset — Action required: reset your password wit▾

Recipients (consented volunteers)

☐ Harshitha G15-CS
<harshithapeddamuregs15@gmail.com>

Notes (optional)

All campaigns

No campaigns yet.

Launch tries GoPhish first and automatically falls back to the Flask SMTP sender if GoPhish is unavailable.

Email Template Selection

Allows selection of phishing email templates.

Round 1 — baseline (before training) ▾

Email template

Password Reset — Action required: reset your password wit▾

Password Reset — Action required: reset your password wit...

Internship Offer — Congratulations — your internship offer ...

Bank Verification — Unusual activity detected — verify your ...

Delivery Failed — Delivery attempt failed — reschedule you...

HR Policy Update — Mandatory: acknowledge the updated HR po...

Notes (optional)

Create as draft

All campaigns

No campaigns yet.

Launch tries GoPhish first and automatically falls back to the Flask SMTP sender if GoPhish is unavailable.

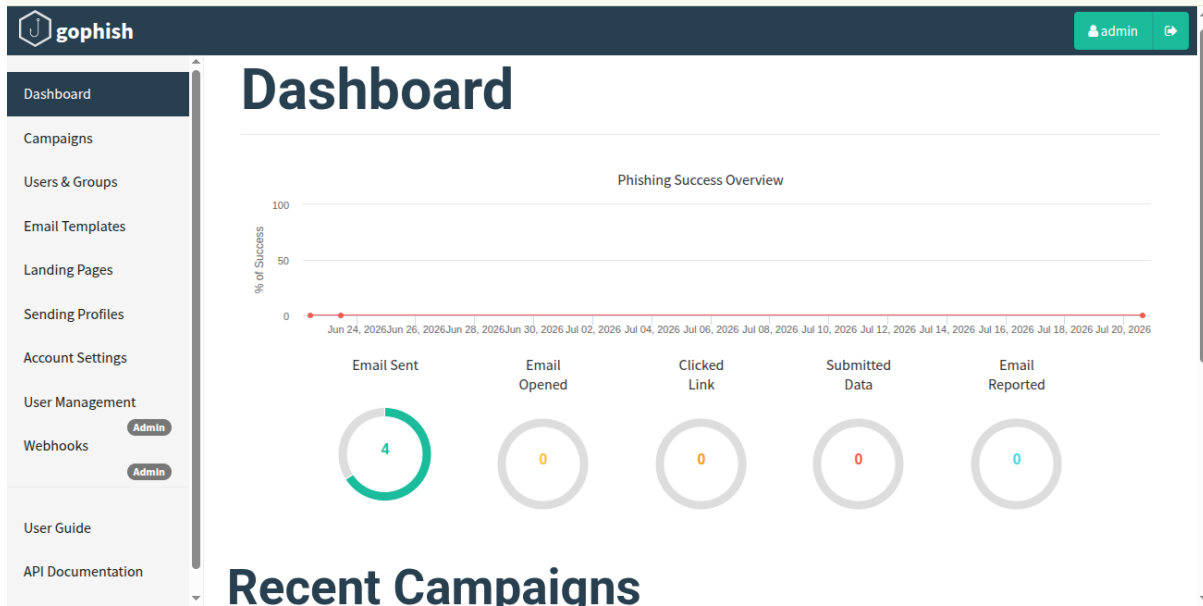
PhishGuard Lab · A consent-based phishing awareness & simulation platform for training and research. Simulated emails only. Passwords are never stored.

GoPhish Integration

Sends phishing campaigns through the GoPhish server.

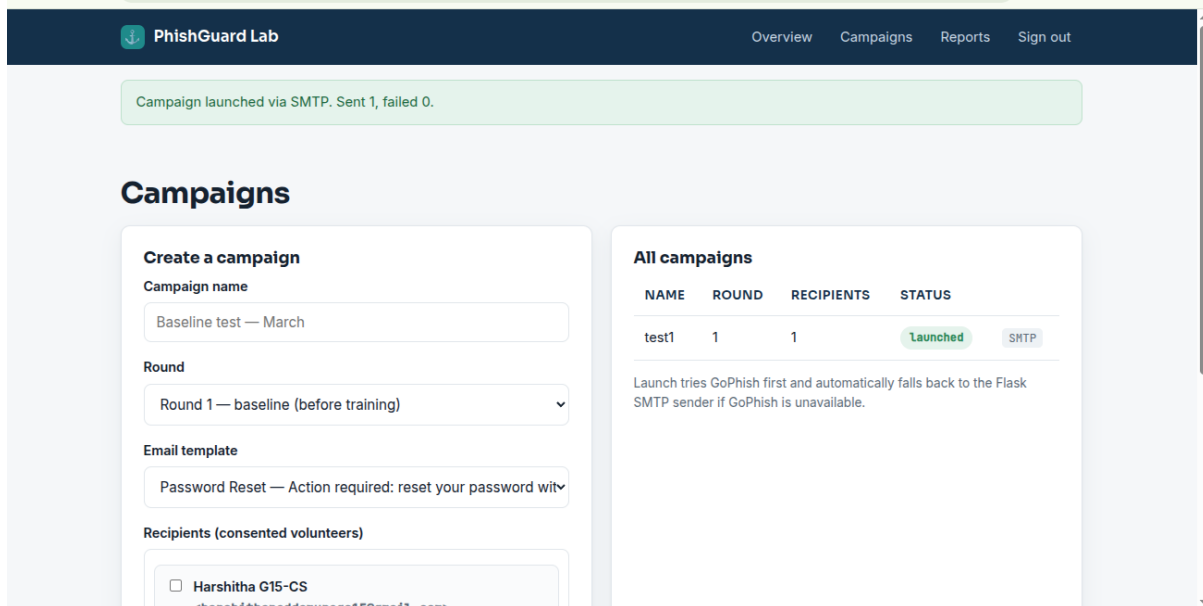


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SMTP Backup

Sends emails when GoPhish is unavailable.

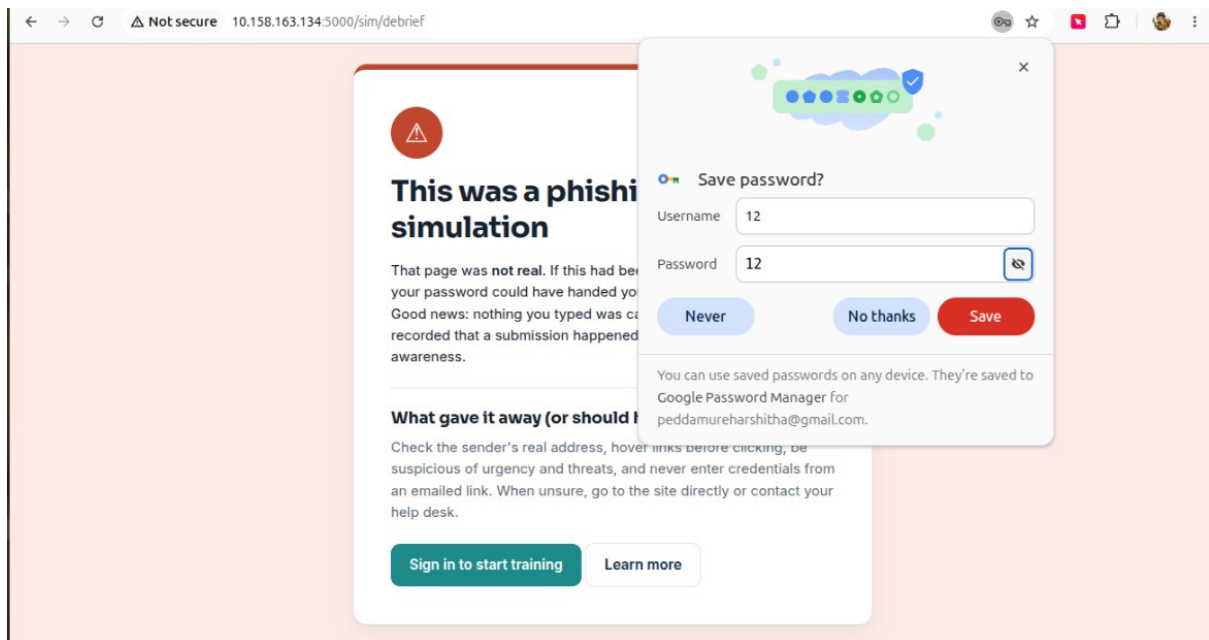
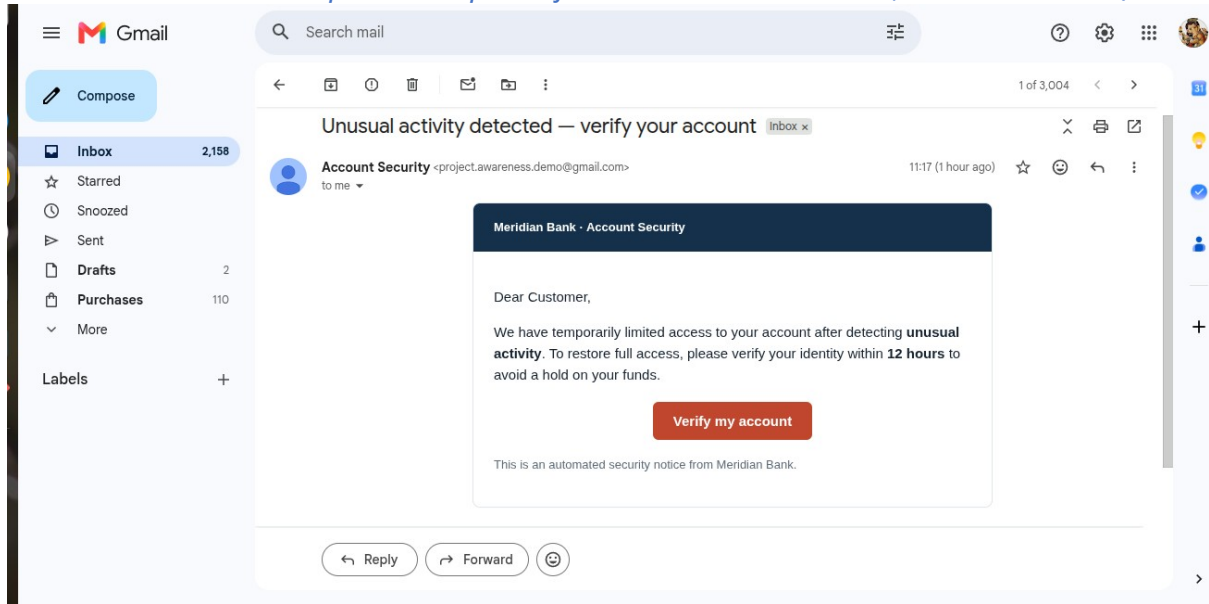


Simulated Landing Page

Displays fake login pages used for awareness training.



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Awareness Training Module

Provides cybersecurity education to participants.



Phishing awareness training

Work through each category, then confirm completion to unlock the quiz.

1 • Email Phishing

most common

Mass emails imitate trusted brands to steal credentials or push malicious links. **Signs:** generic greetings, mismatched sender domains, urgency, and links whose real destination differs from the text. **Defense:** hover links, verify the domain, and go to sites directly rather than via email links.

2 • Spear Phishing

targeted

3 • Smishing (SMS)

4 • Vishing (Voice)

Quiz Module

Assesses participant understanding after training.

 PhishGuard Lab

DashboardTrainingQuizSign out

Training marked complete. Now take the quiz.

Phishing awareness quiz

10 questions. Choose the best answer for each.

1. An email says your account will be locked in 1 hour unless you log in via a link. What is the biggest red flag?

☐ A. The email has a company logo

☐ B. It creates urgency and pressures immediate action

☐ C. It is addressed to you by name

☐ D. It was sent in the morning

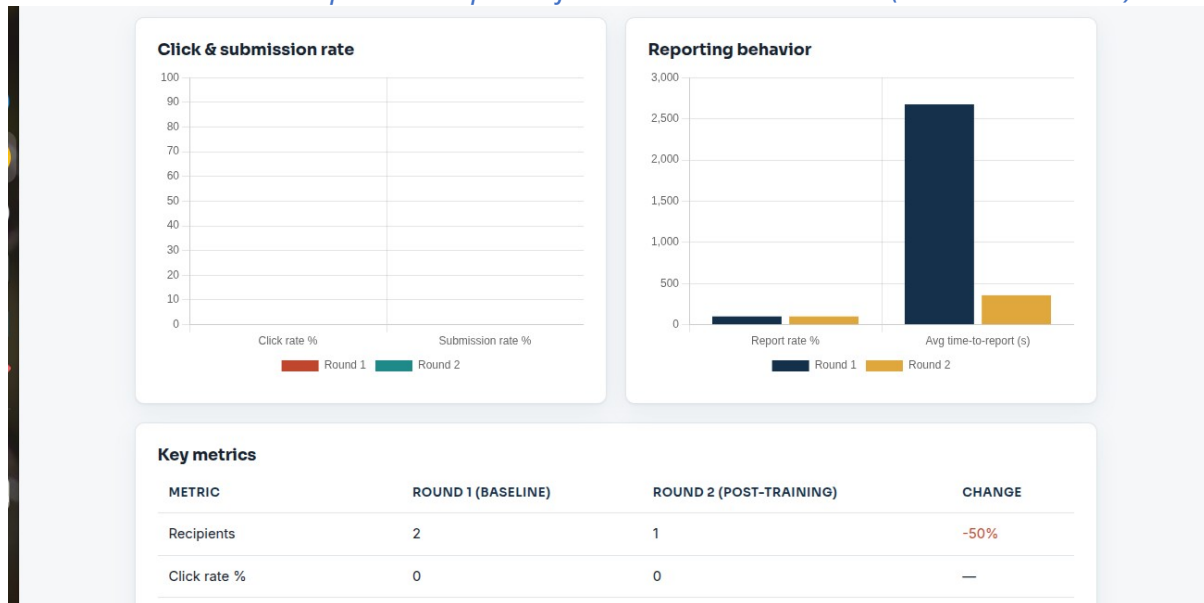
2. Which is the safest way to check where a link really goes?

Reports Dashboard

Displays graphs and performance analytics comparing user awareness before and after training.



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Project GitHub Link

The GitHub repository containing the complete source code, documentation, and project files is provided below.

GitHub Repository: <https://github.com/sure-trust/HARSHITHA-PEDDAMURE-g15-cs>



Learning and Reflection

During the development of Phishing Awareness and Simulation, several new technical and practical concepts were learned. The project provided hands-on experience in building a complete web-based application while understanding the workflow involved in phishing awareness programs.

Table 5.1 Technical Learnings

Area	Learning Outcome
Python & Flask	Learned to develop a full-stack web application using Flask, including routing, templates, sessions, and request handling.
Database Management	Gained experience in designing and managing SQLite databases for volunteers, campaigns, reports, and quiz data.
GoPhish Integration	Learned how phishing simulation platforms work and how GoPhish APIs can be integrated into a Flask application.
SMTP Configuration	Understood email delivery using Gmail SMTP as a backup mechanism for phishing campaigns.
Frontend Development	Improved skills in HTML, CSS, Bootstrap, and JavaScript for developing responsive user interfaces.
Chart.js	Learned to visualize campaign statistics using interactive charts and dashboards.
Authentication	Implemented volunteer and administrator login systems using secure session management.
Ethical Cybersecurity	Understood the importance of participant consent and ethical considerations while conducting phishing awareness simulations.

Working on this project also improved planning, debugging, documentation, and project management skills. Integrating multiple technologies such as Flask, GoPhish, SQLite, SMTP, and Chart.js demonstrated the importance of combining different software components into a unified system.



Conclusion and Future Scope

Conclusion

The system allows administrators to create phishing campaigns using predefined email templates and launch them through **GoPhish** or an **SMTP fallback mechanism**. Participant interactions, including email clicks, simulated credential submissions, phishing reports, and quiz performance, are recorded and analysed through interactive dashboards and graphical reports. The platform also compares participant behaviour before and after awareness activities, enabling administrators to evaluate the effectiveness of the awareness program.

The project successfully achieved its intended objectives by providing a centralized platform for managing phishing awareness campaigns while ensuring ethical participation through informed consent and protecting user privacy by avoiding the storage of real passwords. Overall, **Phishing Awareness and Simulation** demonstrates a practical solution for conducting phishing awareness programs and measuring user awareness through controlled simulations and behavioural analysis.

Future Scope

The current implementation provides all the essential features required for conducting permission-based phishing awareness campaigns. However, a few enhancements can further improve the platform in future versions.

- Enhance the awareness module by including more interactive learning materials and additional phishing scenarios.
- Add more phishing email templates to simulate a wider range of real-world phishing techniques.
- Extend compatibility with other email service providers in addition to the existing GoPhish and SMTP support.